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Education

Moscow State School 57

Physics class, graduated with gold medal
2019–2023

Moscow Institute of Physics and Technology

Bachelor degree
Phystech School of Applied Mathematics and Computer Science

2023–PRESENT

GPA: 9.24/10 (5.0/5)

Achievements

Russian National Physics Olympiad

Twice final stage silver medal
2022, 2023

Candidate to the national physics team for the international Olympiad

2022

Russian National Informatics Olympiad

Twice region stage prize winner
2021, 2023

International Experimental Physics Olympiad (IEPHO)

Bronze medal
2020

Skills

- C / C++ / Go / Haskell / Python
- Assembly x86-64
- Concurrency
- Algorithms and data structures
- Machine learning

Ovakimian Arman

Experience and Projects

Huawei Intern

JUL 2025 – PRESENT

Intern at Huawei on the DSP compilers team, identifying performance and code-size issues in assembly code and implementing optimizations to address them. Compiler is based on LLVM.

Huawei Open Days

AUG 2024

Applied Retrieval-Augmented Generation (RAG) techniques to improve code generation quality.

X86-64 Compiler for my own programming language

APR 2024 – MAY 2024

GH link: <https://github.com/d3clane/Compiler>

My own esoteric programming language with compiler to executable x86-64 elf64 file. Implemented frontend, middle-end and backend.

LLVM Pass

APR 2025

GH link: https://github.com/d3clane/llvm_pass

Developed an LLVM instrumentation pass that inserts logging calls to collect dynamic execution data, including common control-flow paths and dynamically allocated object usage, then maps the collected data into a graph representation.

Continuous Exchange MVP

JAN 2026 – MAY 2026

Co-developed a continuous-exchange MVP inspired by Kyle and Lee's *Toward a Fully Continuous Exchange*, with features including discrete-to-continuous order-book conversion, order matching/clearing, external liquidity integration, and demo-mode execution on external venues.

Concurrency Course

FEB 2025 – MAY 2025

1. Implemented synchronization primitives (mutex, condvar, spinlock, semaphore, waitgroup, etc.) using atomic operations only. Designed FIFO spinlock that avoids false sharing and the thundering herd problem.
2. Built a work stealing thread pool and fibers.
3. Developed synchronization primitives for fibers (mutex, event, waitgroup, channel, etc.).
4. Implemented futures — a functional approach to concurrency.

Languages

- English (B2)
- Russian (Native)

Graphics editor

NOV 2024 – DEC 2024

GH link: <https://github.com/d3clane/MyPhotoshop>

Co-designed a standard plugin API for a student graphics editor, then developed my own editor implementation and plugins based on the API, including brush, eraser, Gaussian blur, and brightness filters.

Optimizing hash table using asm

APR 2024

GH link: <https://github.com/d3clane/HashTable>

Optimizing hash table requests time performance using asm. Three types of optimizations are used:

1. SIMD instructions
 2. Inline asm
 3. Linking with another file with asm code
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Hacking my friend's program (crackme)

FEB 2024

GH link: <https://github.com/d3clane/DosHack>

Finding vulnerabilities in my friend's program and exploiting them.